

Probabilistic Inverse Semigroups

as Markov Chains on Partial Symmetries

A Short Note — Corrected Final Version

Abstract

We introduce the notion of a probabilistic inverse semigroup: a finite inverse semigroup equipped with a locally normalized probability distribution on allowed multipliers, invariant under the inversion map. The structure defines a right-handed Markov chain whose states are the semigroup elements and whose transitions correspond to right multiplication by a randomly chosen multiplier. We show that this chain is dual, via the semigroup involution, to a left-handed chain derived from the same data. The duality yields a path-symmetry theorem, and a worked example illustrates the full consistency of the construction.

1. Introduction

Inverse semigroups capture partial symmetries: every element x has a unique generalised inverse x^{-1} satisfying $x \cdot x^{-1} \cdot x = x$ and $x^{-1} \cdot x \cdot x^{-1} = x^{-1}$. They appear naturally in the description of reversible computations with partial information, quantum error correction, and the geometry of groupoids.

This note introduces a probabilistic layer on top of an inverse semigroup, turning it into a Markov chain that respects the algebraic involution.

Earlier naive attempts led to normalisation inconsistencies by conflating single-step probabilities with path probabilities. Here we adopt a clear separation of levels: the basic object is a choice function on allowed multipliers, the transition kernel is built by composition, and path probabilities follow the standard Chapman–Kolmogorov calculus.

The only algebraic–probabilistic link imposed is inverse symmetry (Equation I), which is remarkably compatible with the semigroup structure and leads to an elegant duality.

2. Definitions

Definition 1 (Partial inverse semigroup).

A partial inverse semigroup is a set S equipped with a partially defined binary operation

$\cdot : D \rightarrow S$ (where $D \subseteq S \times S$) and a total unary operation $^{-1} : S \rightarrow S$ such that:

- (i) For all x , $(x, x^{-1}) \in D$ and $x \cdot x^{-1} \cdot x = x$ whenever defined;
- (ii) $x^{-1} \cdot x \cdot x^{-1} = x^{-1}$;
- (iii) $(x^{-1})^{-1} = x$;
- (iv) Idempotents commute.

All structures considered are finite.

Definition 2 (Domain involution closure).

S satisfies the domain involution closure if:

$$\forall x, y \in S : (x, y) \in D \Leftrightarrow (y^{-1}, x^{-1}) \in D$$

This condition is automatic in many natural examples (e.g. symmetric inverse semigroup I_X).

It ensures that left- and right-handed probabilistic chains are defined on the same domain.

Definition 3 (Probabilistic choice function).

Let $(S, \cdot, ^{-1})$ satisfy (D). A map $P : D \rightarrow (0,1]$ is a probabilistic choice function if:

- (N) Local normalisation: for every $x \in S$, the set $\{y \mid (x, y) \in D\}$ is non-empty and $\sum_{\{y:(x,y) \in D\}} P(x \rightarrow y) = 1$
- (I) Inverse symmetry: for all $(x, y) \in D$,
 $P(x \rightarrow y) = P(x^{-1} \rightarrow y^{-1})$

A tuple $(S, \cdot, ^{-1}, P)$ satisfying (D), (N) and (I) is a probabilistic inverse semigroup.

2. Definitions (continued)

Definition 4 (Right-handed Markov chain).

The right transition kernel $K_R : S \times S \rightarrow [0,1]$ is defined by:

$$K_R(x, z) = \sum_{\{y:(x,y) \in D, x \cdot y = z\}} P(x \rightarrow y)$$

It is a stochastic matrix because of (N). Path probability:

$$\mathbb{P}_R(x_0 \rightarrow^{y^1} x_1 \rightarrow^{y^2} \dots \rightarrow^{y^n} x_n) = \prod_{i=1}^n P(x_{i-1} \rightarrow y_i)$$

where $x_i = x_{i-1} \cdot y_i$. The multi-step kernel $K_R^{(n)}$ obeys Chapman–Kolmogorov.

Definition 5 (Left-handed chain).

By (D), for every x the set $\{y \mid (y,x) \in D\}$ is in bijection with

$\{w \mid (x^{-1},w) \in D\}$ via $w = y^{-1}$. The left-multiplier choice probability is:

$$P_L(x \rightarrow y) := P(x^{-1} \rightarrow y^{-1})$$

The left transition kernel is:

$$K_L(x, z) = \sum_{\{y:(y,x) \in D, y \cdot x = z\}} P_L(x \rightarrow y)$$

Local normalisation for K_L follows from (N) and (D):

$$\begin{aligned} \sum_z K_L(x,z) &= \sum_{\{y:(y,x) \in D\}} P(x^{-1} \rightarrow y^{-1}) \\ &= \sum_{\{w:(x^{-1},w) \in D\}} P(x^{-1} \rightarrow w) = 1 \end{aligned}$$

3. The Duality Theorem

Theorem 1 (Inversion duality).

For all $x, z \in S$:

$$K_L(x, z) = K_R(x^{-1}, z^{-1})$$

Proof. Using the bijection $w = y^{-1}$ guaranteed by (D), and the identity $(y \cdot x)^{-1} = x^{-1} \cdot y^{-1}$ valid in any inverse semigroup, we have:

$$\begin{aligned} K_L(x, z) &= \sum_{\{y: (y, x) \in D, y \cdot x = z\}} P(x^{-1} \rightarrow y^{-1}) \\ &= \sum_{\{w: (x^{-1}, w) \in D, x^{-1} \cdot w = z^{-1}\}} P(x^{-1} \rightarrow w) \\ &= K_R(x^{-1}, z^{-1}). \end{aligned}$$

QED.

Corollary 2 (Path symmetry).

Any left-handed path $x_0 \rightarrow^{y^1}_L x_1 \rightarrow^{y^2}_L \dots \rightarrow^{y^n}_L x_n$ has probability exactly equal to the right-handed path:

$$x_0^{-1} \rightarrow^{y^{1-1}}_R x_1^{-1} \rightarrow^{y^{2-1}}_R \dots \rightarrow^{y^{n-1}}_R x_n^{-1}$$

because term-by-term: $P(x_{i-1}^{-1} \rightarrow y_i^{-1}) = P(x_{i-1} \rightarrow y_i)$ by (I), and

$$x_{i-1}^{-1} \cdot y_i^{-1} = (y_i \cdot x_{i-1})^{-1} = x_i^{-1}.$$

4. A Concrete Example

Let $S = \{0, e, a, a^*\}$ with partial multiplication (undefined entries marked '—'):

·	0	e	a	a^*
0	0	0	0	0
e	0	e	a	a^*
a	0	a	—	e
a^*	0	a^*	e	—

Inverses: $0^{-1}=0$, $e^{-1}=e$, $a^{-1}=a^*$, $(a^*)^{-1}=a$.

Condition (D) holds: e.g. $(a, a^*) \in D$ and $(a, a^*)^{-1} = (a^*, a) \in D$.

Probabilities $P(x \rightarrow y)$:

From e: $P(e \rightarrow 0)=0.3$, $P(e \rightarrow e)=0.1$, $P(e \rightarrow a)=0.3$, $P(e \rightarrow a^*)=0.3$ (sum=1)

From a: $P(a \rightarrow 0)=0.3$, $P(a \rightarrow e)=0.2$, $P(a \rightarrow a^*)=0.5$ (sum=1)

From a^* : $P(a^* \rightarrow 0)=0.3$, $P(a^* \rightarrow e)=0.2$, $P(a^* \rightarrow a)=0.5$ (sum=1)

From 0: $P(0 \rightarrow 0)=0.25$, $P(0 \rightarrow e)=0.25$, $P(0 \rightarrow a)=0.25$, $P(0 \rightarrow a^*)=0.25$ (sum=1)

Inverse symmetry (I) verified:

$$P(a \rightarrow a^*) = 0.5 = P(a^* \rightarrow a)$$

$$P(a \rightarrow e) = 0.2 = P(a^* \rightarrow e)$$

4. Example (continued)

Right kernel K_R :

From e : to 0 : 0.3, to e : 0.1, to a : 0.3, to a^* : 0.3 (sum=1)

From a : to 0 : 0.3, to a : 0.2 (via e), to e : 0.5 (via a^*) (sum=1)

From a^* : to 0 : 0.3, to a^* : 0.2 (via e), to e : 0.5 (via a) (sum=1)

From 0 : to 0 : 1.0 (absorbing: $0 \cdot y = 0$ for all y) (sum=1)

Checking Theorem 1:

$$K_L(a, e) = P_L(a \rightarrow a^*) = P(a^* \rightarrow a) = 0.5$$

(since $a^* \cdot a = e$ is the only left multiplier yielding e from a)

$$K_R(a^{-1}, e^{-1}) = K_R(a^*, e) = P(a^* \rightarrow a) = 0.5$$

(since $a^* \cdot a = e$ is the only right multiplier yielding e from a^*)

The two values match perfectly: $K_L(a, e) = K_R(a^*, e) = 0.5$.

5. Discussion and Outlook

The construction is fully consistent and provides a minimal framework in which probabilistic and partial-inverse algebraic structures coexist without normalisation conflicts.

Directions:

Stationary distributions and irreducibility. The zero element is absorbing.

Communicating classes of K_R relate to Green relations of the semigroup.

From finite to infinite. Requires summability conditions on allowed multipliers.

Quantum extensions. Correctable operations in quantum error correction form an inverse semigroup. The duality theorem relates forward and backward (dual) error propagation.

Symmetry-driven design. Given any finite inverse semigroup satisfying (D), construct symmetric P by averaging an arbitrary distribution with its image under inversion.

This note lays the algebraic groundwork. A fuller treatment with large deviations, entropy, and cryptographic applications will appear elsewhere.

References

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